

Instruction Format

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Instruction format

Compute Memory traffic, total memory for encoding and storing code that implements the expression evaluation $a = (b + c) * d - e$. Assume that the opcode occupy one byte, addresses occupy three bytes, data values also occupy three bytes and 3 byte word length

Evaluate $a = (b+c)*d - e$

3-Address

add a, b, c $a \leftarrow b+c$

mpy a, a, d $a \leftarrow a*d$

sub a, a, e $a \leftarrow a-e$

2-Address

mov a, b $a \leftarrow b$

add a, c $a \leftarrow a+c$

mpy a, d $a \leftarrow a*d$

sub a, e $a \leftarrow a-e$

1-Address

load b $Acc \leftarrow b$

add c $Acc \leftarrow Acc+c$

mpy d $Acc \leftarrow Acc*d$

sub e $Acc \leftarrow Acc-e$

store a $a \leftarrow Acc$

0-Address

push b

push c

add

push d

mpy

push e

sub

pop a

add a, b, c
 mpy a, a, d
 sub a, a, e

$a \leftarrow b+c$
 $a \leftarrow a*d$
 $a \leftarrow a-e$

Memory to Store	Memory to encode	M/As to Fetch	M/As to Execute	Memory Traffic
$4*3=12$	$1+(3*3)=10$	4	3	$4+3=7$
$4*3=12$	$1+(3*3)=10$	4	3	$4+3=7$
$4*3=12$	$1+(3*3)=10$	4	3	$4+3=7$
36	30	12	9	21

Mov a,b
 Add a,c
 mpy a, d
 Sub a,e

$a \leftarrow b$
 $a \leftarrow a+c$
 $a \leftarrow a*d$
 $a \leftarrow a-e$

Memory to Store	Memory to encode	M/As to Fetch	M/As to Execute	Memory Traffic
$3*3=9$	$1+(2*3)=7$	3	2	5
$3*3=9$	$1+(2*3)=7$	3	3	6
$3*3=9$	$1+(2*3)=7$	3	3	6
$3*3=9$	$1+(2*3)=7$	3	3	6
36	28	12	11	23

load b	$\text{Acc} \leftarrow b$
add c	$\text{Acc} \leftarrow \text{Acc} + c$
mpy d	$\text{Acc} \leftarrow \text{Acc} * d$
sub e	$\text{Acc} \leftarrow \text{Acc} - e$
store a	$a \leftarrow \text{Acc}$

Memory to Store	Memory to encode	M/As to Fetch	M/As to Execute	Memory Traffic
2*3=6	1+(1*3)=4	2	1	2+1=3
2*3=6	1+(1*3)=4	2	1	2+1=3
2*3=6	1+(1*3)=4	2	1	2+1=3
2*3=6	1+(1*3)=4	2	1	2+1=3
2*3=6	1+(1*3)=4	2	1	2+1=3
2*3=6	1+(1*3)=4	2	1	2+1=3
30	20	10	5	15
2*3=6	1+(1*3)=4	2	1	3
2*3=6	1+(1*3)=4	2	1	3
3	1	1	0	1
2*3=6	1+(1*3)=4	2	1	3
3	1	1	0	1
2*3=6	1+(1*3)=4	2	1	3
3	1	1	0	1
2*3=6	1+(1*3)=4	2	1	3
39	23	13	5	15

Summary of 3-,2-,1-, and 0-Addresss instruction programming

Instruction Format	Memory to Store	Memory to Encode	M/As to Fetch	M/As to Execute	Memory Traffic
4-Address	45	39	24	9	33
3-Address	36	30	12	9	21
2-Address	36	28	12	11	23
1-Address	30	20	10	5	15
0-Address	39	23	13	5	15

Instruction format

Compute Memory traffic, total memory for encoding and storing code that implements the expression evaluation $A = (B - C) * D$. Assume that the opcode occupy one byte, addresses occupy two bytes, data values also occupy two bytes and 1 byte word length